FUTURE IMPACT OF AI

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Introduction

PROBLEM DEFINITION

The AI race is rapidly influencing and shaping how we work but the impact is highly uneven.

Companies and industries face a hurdle while working towards **AI adoption** because they lack insight on how they are most likely and least like to to be affected

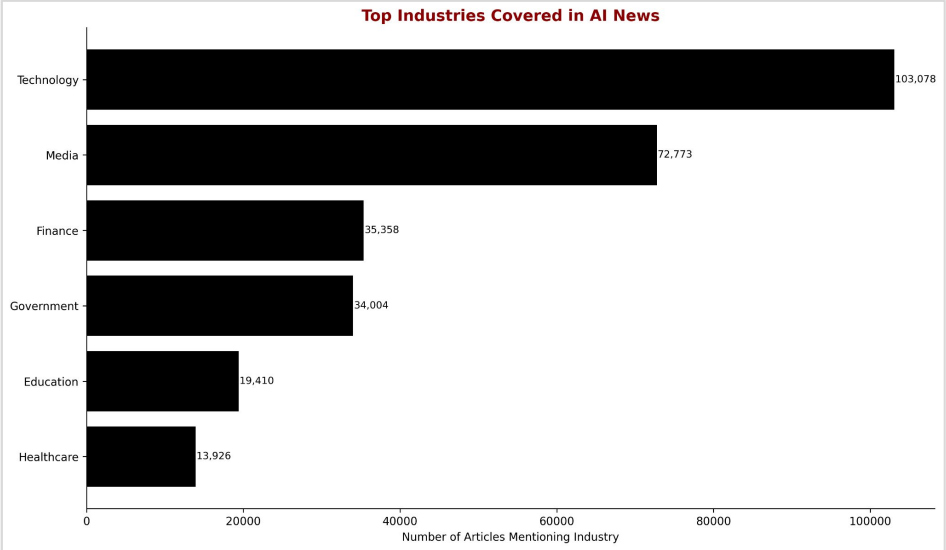
This project uses data from News Articles to understand: **WHO**; which industries and companies are most exposed to AI disruption, **HOW**; whether that exposure is viewed positively or negatively, and **WHAT** separates successful AI adoption from failure.

Industry Analysis

- **Technology is the Leader:** With over **31%** of the classified articles, this sector is the most frequently covered industry in AI news, indicating its central role and significant impact.
- **Media Sector Shows High Engagement:** The Media industry also features prominently, contributing **22%**, suggesting that AI's influence on content creation, distribution, and consumption is a major topic.
- **Finance and Government**: Both Finance and Government, **10%** each are heavily discussed, pointing to AI's transformative potential and regulatory challenges in these critical sectors.
- **Education and Healthcare are Adapting:** With **19,000+** and **13,000+** articles respectively, Education and Healthcare are clearly grappling with AI integration, from new learning tools to diagnostic advancements.

Top Four Classified Industries

31%	Technology	10%	Finance
22%	Media	10%	Government



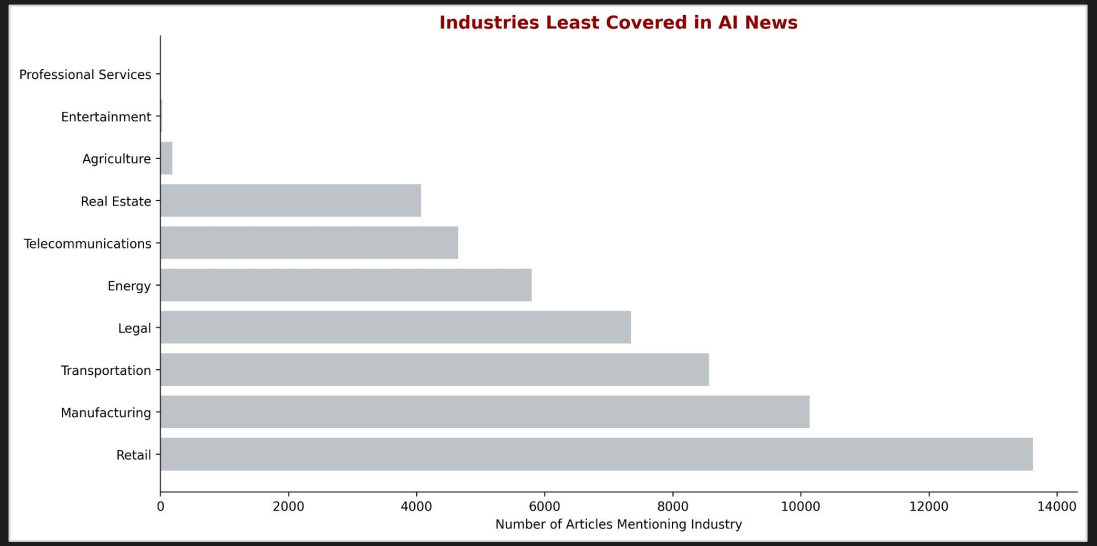
Least Mentioned Industries

Three industries appear at the bottom of the corpus by article volume:

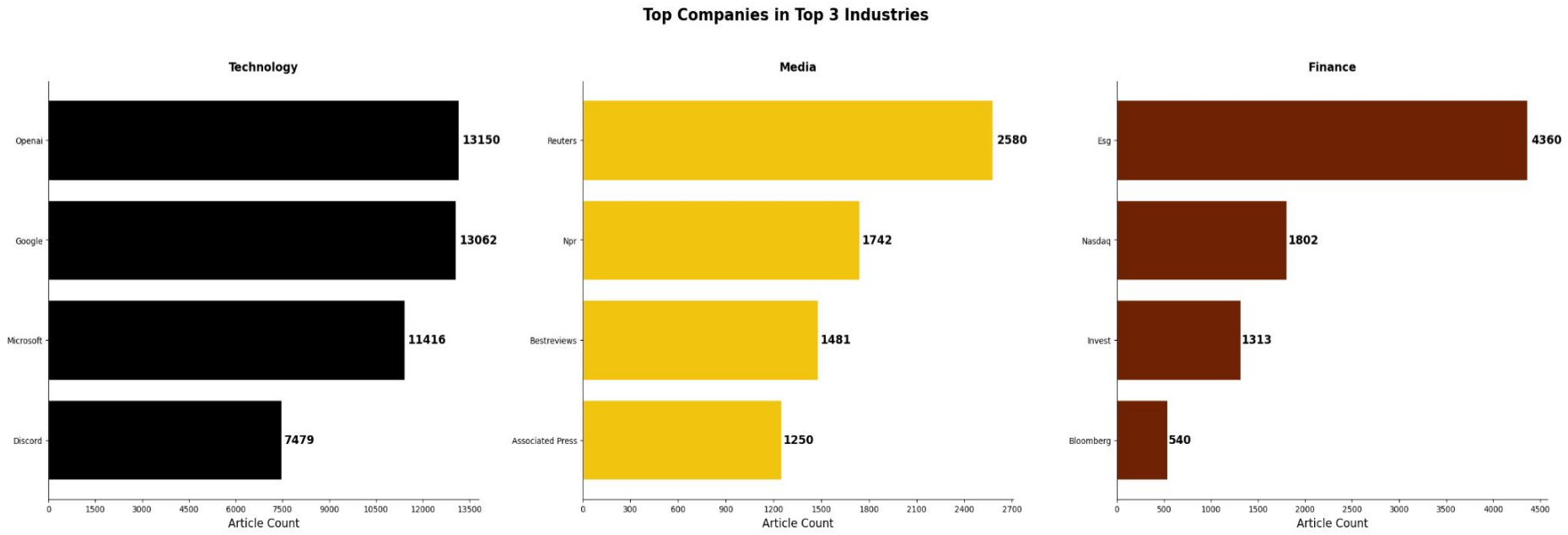
- Professional Services (16 articles)
- Entertainment (36 articles)
- Agriculture (197 articles)

These sectors are generating significantly less public discourse around AI.

Technology and Professional Services, the most and least represented industries, have a coverage gap of roughly **6,400x**

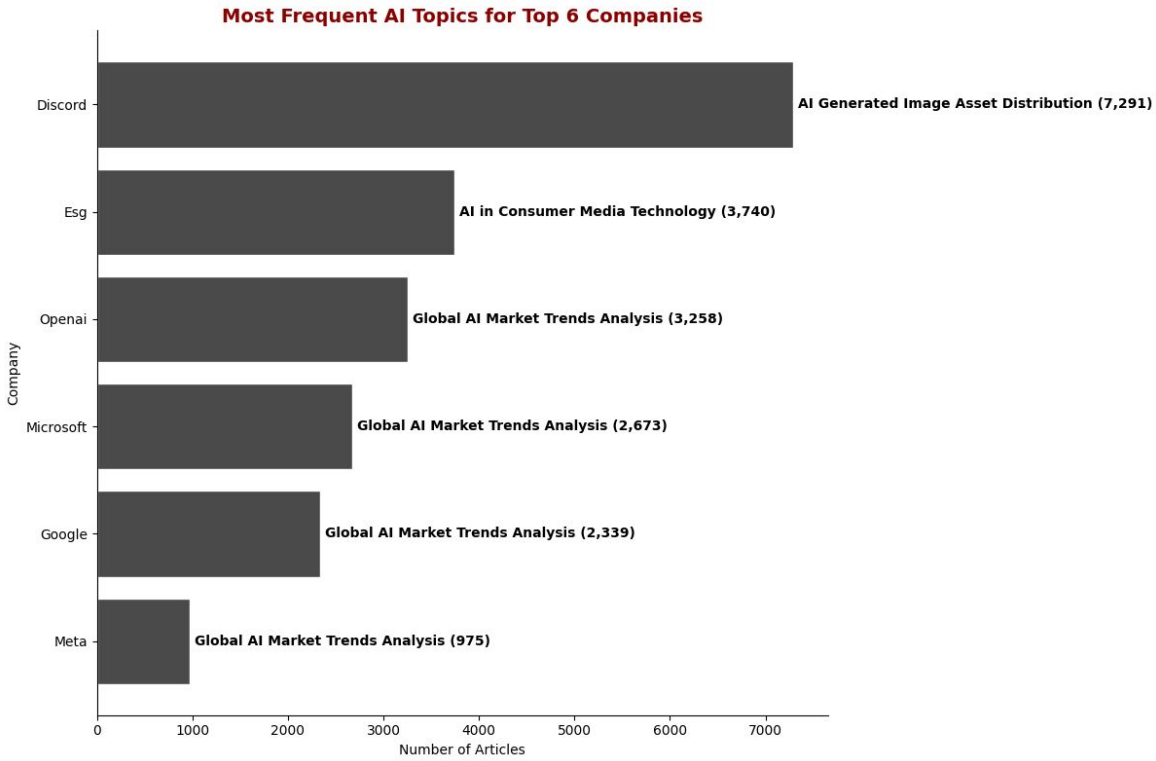


Top Companies



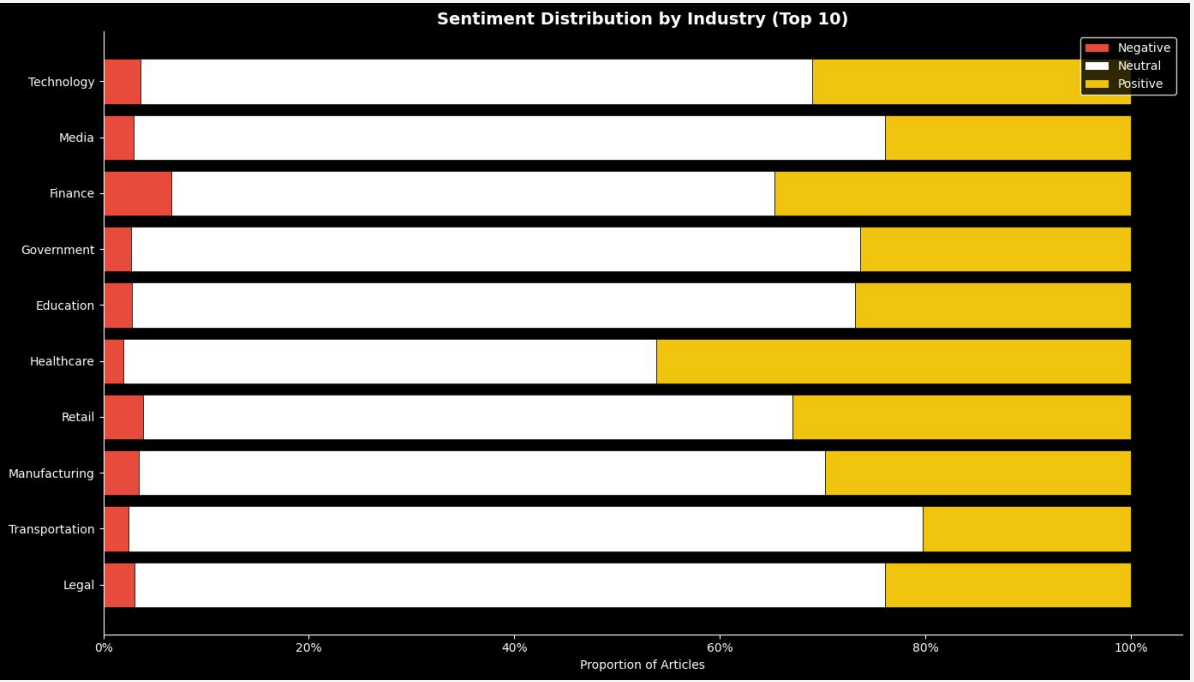
Technology is the largest industry in the corpus at 103,000 articles, it also 100% dominates **top 6** overall company-level mention counts sitting at the center of the AI narrative.

Technologies Impacting Companies



- **Discord:** Its most frequent AI topic is **AI Generated Image Asset Distribution**, suggesting its public perception is heavily influenced by the use and distribution of AI-generated visual content.
- **Openai:** Its most frequent AI topic is **Global AI Market Trends Analysis**. This reflects its central role in shaping and driving the overall AI market, with public perception keenly observing its influence on broader industry trends and economic impact.
- **Microsoft:** Its most frequent AI topic is **Global AI Market Trends Analysis**, similar to OpenAI, emphasizing its significant influence and involvement in the strategic direction and economic aspects of the AI industry.
- **Google:** Its most frequent AI topic is **Global AI Market Trends Analysis**, also aligning with OpenAI and Microsoft, highlighting its position as a major player whose AI developments are perceived as key indicators of global AI market dynamics.
- **Meta:** Its most frequent AI topic is **Global AI Market Trends Analysis**, enforcing the idea that the public views its AI endeavors through the lens of overall market shifts and competitive landscape.

Industry Sentiment Analysis positively or negatively impacted



Neutral Dominance

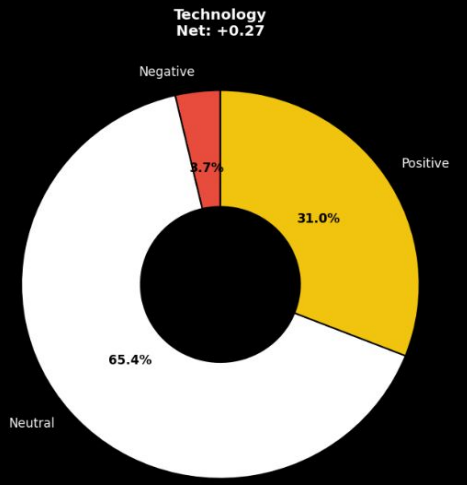
Across all industries neutral sentiment accounts for the majority of coverage ranging from **44%** in Healthcare to **77%** in Transportation.

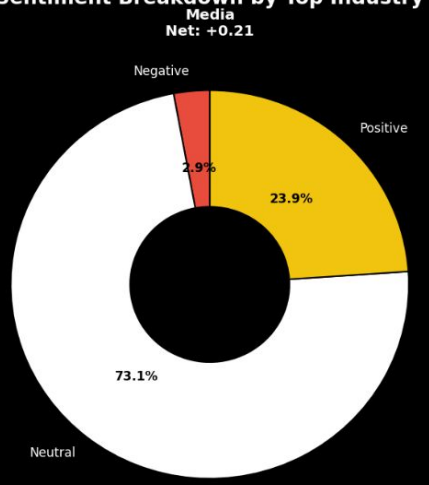
Net sentiment scores therefore reflect the balance of the non-neutral coverage rather than absolute enthusiasm or concern.

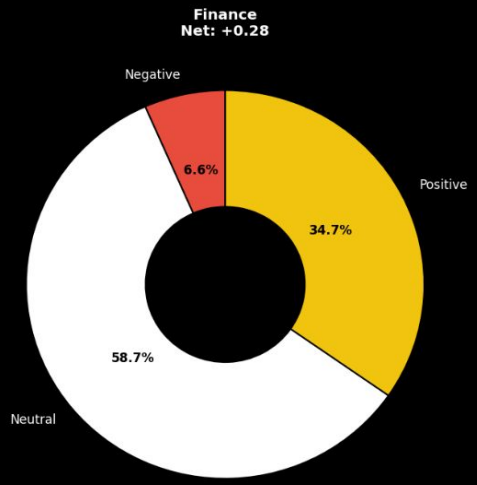
Positive Skew

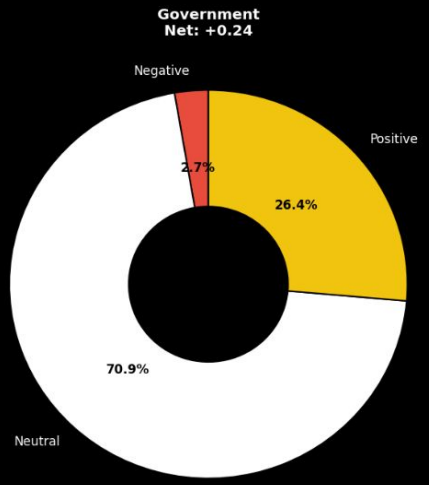
AI coverage skews net positive across all 16 identified industries however the degree of positivity varies.

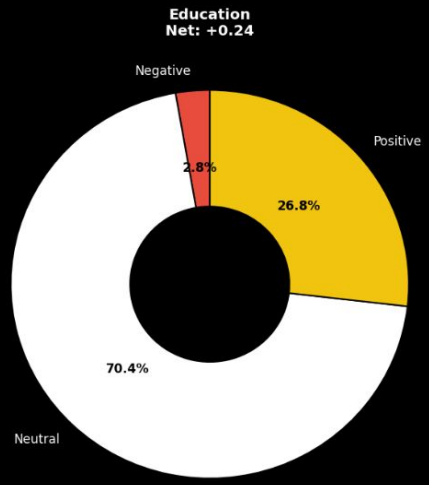
Sentiment Breakdown by Top Industry

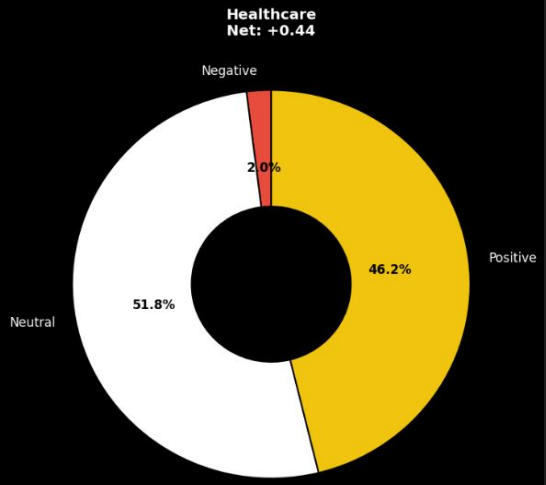






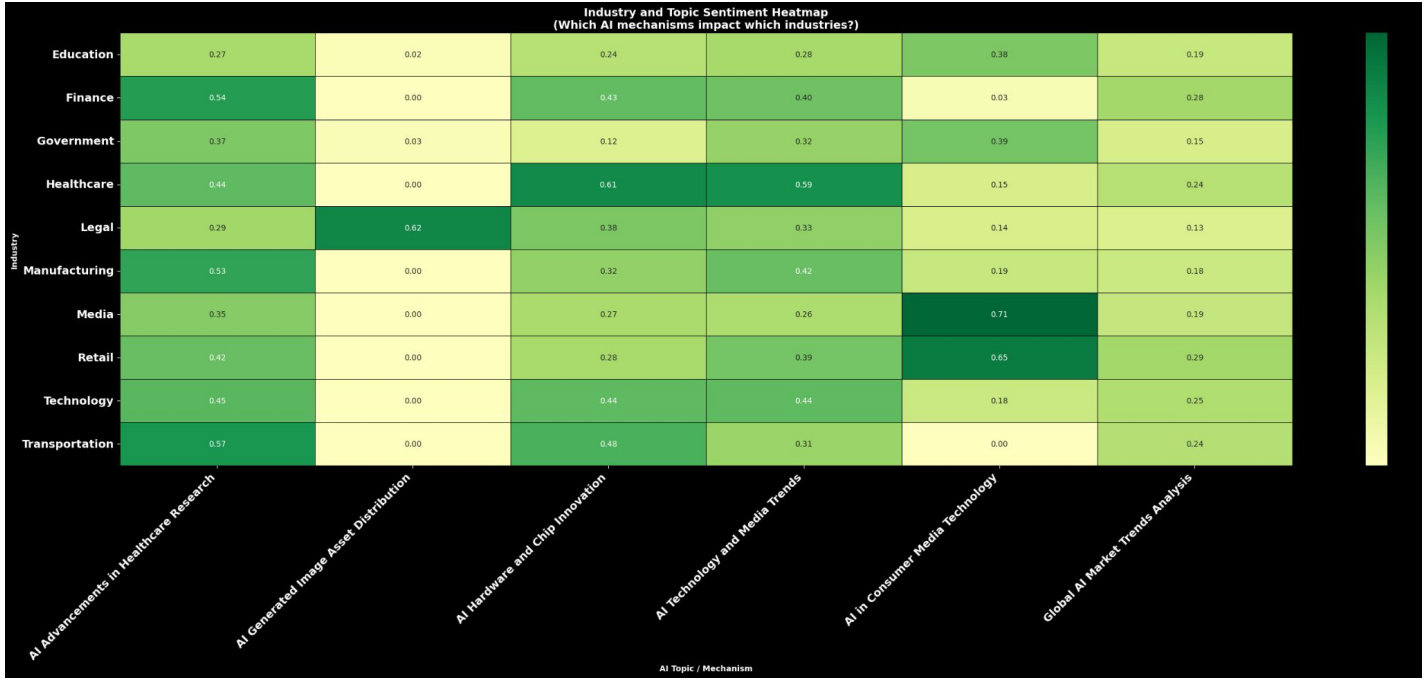


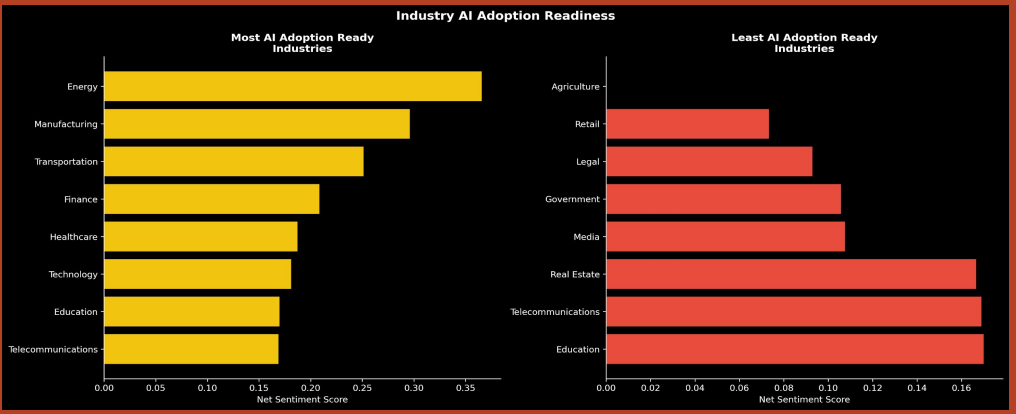


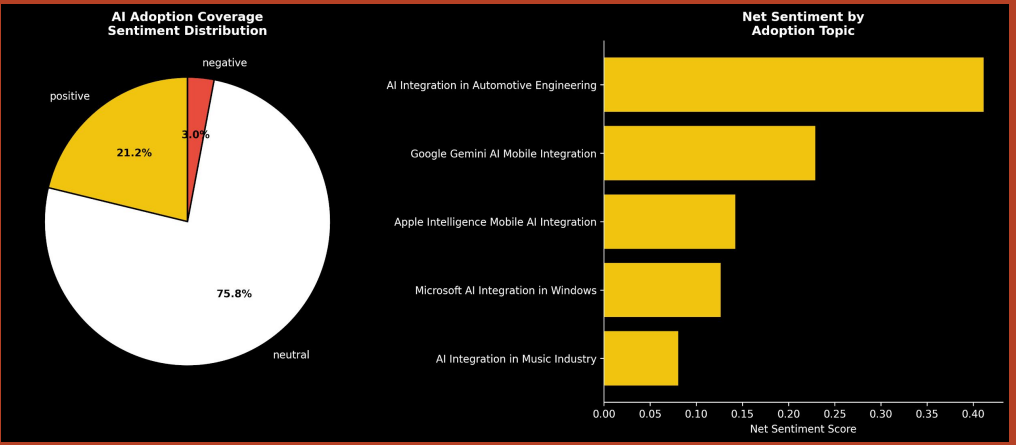


AI Mechanisms Driving Industry Impact by what means

Industry-topic sentiment patterns reveals that AI impact is primarily driven through **augmentation** rather than replacement. **AI in Healthcare Research** shows the strongest and most consistent positive sentiment signal across the widest range of industries consistent with AI being used to enhance existing workflows such as diagnostics, asset management, and predictive analytics. **AI Hardware and Chip Innovation** registers strong positive sentiment particularly in Technology, Transportation, and industries where infrastructure investment in AI compute is a central narrative. **AI in Consumer Media Technology** drives the strongest sentiment signal in both Media and Retail, where the dominant mechanism is **workflow redesign**. In the Legal sector, **AI Generated Image and Asset Distribution** is the dominant mechanism, framing AI's impact primarily through **regulatory and compliance pressure** rather than operational efficiency. *Cells with near-zero values across the heatmap indicate industry-topic combinations where coverage is too sparse to draw directional conclusions i.e.* ***inconclusive rather than unimpacted.***







AI Adaption

- **Industry Sentiment Rankings:** The Energy industry holds the highest net sentiment score for AI adoption readiness (approximately 0.37), while Agriculture and Retail are listed among the least ready, with scores below 0.08.
- **Sector Overlap:** Education and Telecommunications are right in the middle indicating a divergent sentiment within those specific sectors.
- **Topic-Specific Performance:** Among specific adoption topics, **AI Integration in Automotive Engineering** has the highest net sentiment score (exceeding 0.40), whereas **AI Integration in the Music Industry** has a low positive sentiment score (below 0.10).

Successful vs Unsuccessful Adaptation

Successful

High Precision & Standardized Data

Based on the data, industries that rely on high precision, such as Energy and Manufacturing, are the most "ready" for AI. These sectors typically operate with structured, time-stamped, and "clean" datasets (like sensor logs or grid load data) which allow AI models to produce highly accurate and actionable results, leading to the highest net sentiment scores.

Infrastructure Ready Environments

Sectors like Technology and Finance show higher readiness because they have the pre-existing digital "plumbing" to integrate AI seamlessly into their existing workflows without needing a total overhaul of their physical systems.

Unsuccessful

Inadequate Capital/Infrastructure

Industries like Retail and Agriculture appear in the Least Ready. When an industry lacks the capital to invest in high-grade data cleaning or infrastructure, AI adaption is limited.

Creativity & Subjectivity

AI adoption is less successful in sectors where value is derived from subjective human experiences and emotions. This is reflected in the Music Industry having the lowest sentiment score, as the data suggests users remain skeptical of AI's ability to replicate artistic & emotional depth.

Article Clean-up & Filtering

Data Overview

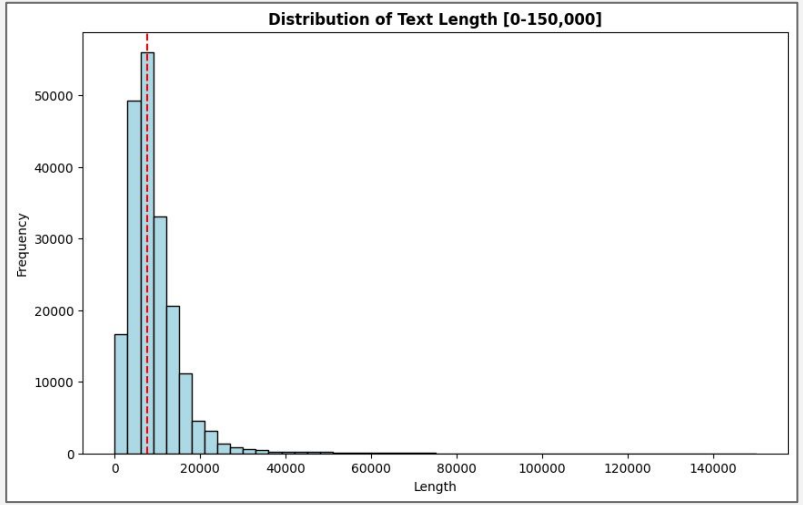
The raw corpus contained **199,989** news articles across **5 columns**: URL, date, language, title, and text. All columns were fully populated with zero null values. Articles spanned multiple years (**2022-2026**) with a median length of **7,621 characters** and a maximum of **580,772 characters**, indicating significant variation in article depth and quality.

Deduplication

494 exact duplicate articles were identified by text content and removed using a keep-first strategy. An additional **42,964 articles** shared duplicate titles. These were reviewed and retained since identical headlines can appear across different publications covering the same story with distinct body content.

Length Filtering & Quality Cleaning

Articles under **200 characters** were removed as structurally insufficient to carry meaningful content. A multi-stage cleaning pipeline was applied using **BeautifulSoup** to decompose HTML structural tags, nav, footer, script, style, header followed by r**egex pattern matching** to remove UI artifacts, repeated separators, and formatting symbols. Cleaned text was **whitespace-normalized** to produce the final text_clean column used in all downstream analysis.



Article Clean-up and Filtering

Final Corpus: 196,629

199,989 articles → 199,495 articles → 198,610 articles → 198,412 articles → 196,629 articles

Deduplication:
Removed rows with duplicate text

Length Filter:
Articles under 200 characters were removed

Language Filter:
Language detection was performed using **pycld2** for its speed advantage at scale. Non-English articles were removed, retaining only English and unknown-language articles.

Relevance Filter:
A keyword-based relevance filter was applied to retain only articles substantively related to AI, machine learning, and data science.

Topic Detection

→ Why BERTopic

BERTopic discovers technology themes directly from the language patterns in the 196,629 articles, surfacing both established technologies and emerging ones that a manual list would miss. Topics are defined by their most discriminative terms, making them interpretable as technology categories.

→ BERTopic Model Configuration

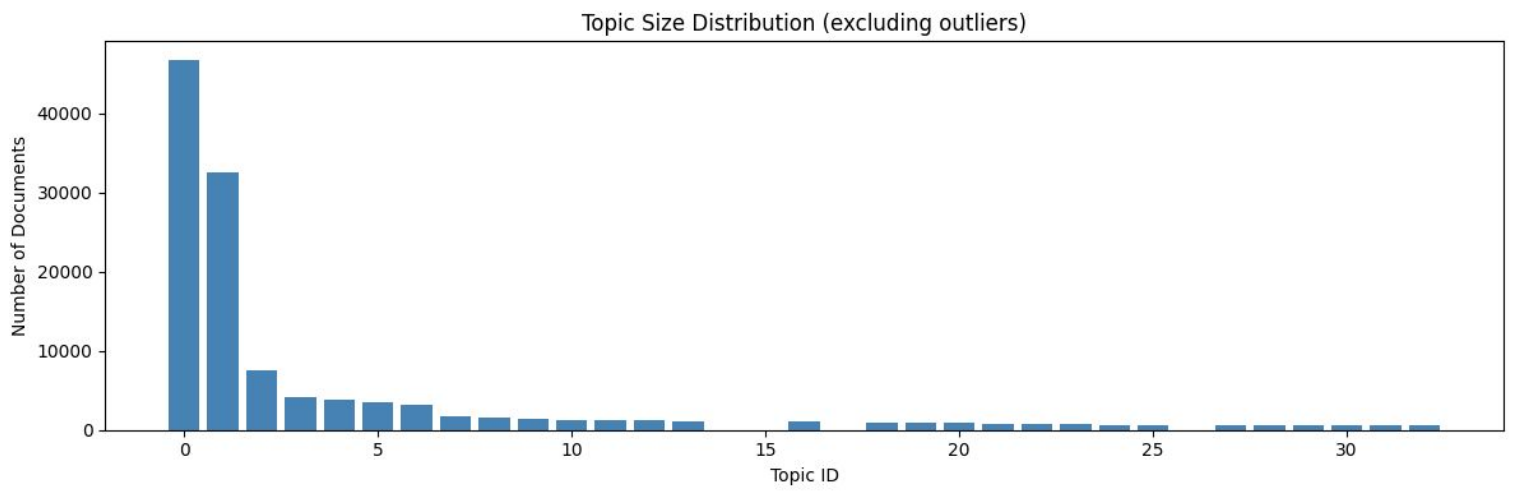
Component	Choice	Rationale
Embedding Model	all-MiniLM-L6-v2	Fast, high quality sentence embeddings
Dimensionality Reduction	UMAP	Preserves local and global structure
Clustering	HDBSCAN	Handles variable density clusters, noise robust
Vectorizer	CountVectorizer (1-2 ngrams)	Captures multi-word technology terms
Topic Reduction	Auto	Data-driven, avoids over-specification
Corpus	196,629 articles	Full corpus, no sampling

Topic Detection

BERTopic identified 33 topics from the corpus,excluding outliers, each labeled using Gemini API to produce a 4-6 word human-readable description. Five topics were excluded as noise:

- **Topic -1**: HDBSCAN outliers, articles insufficiently similar to any cluster
- **Topic 14**: German-language cluster, non-English content that passed the language filter
- **Topics 15 & 17**: Stock ticker and financial instrument listing pages, structured data pages with no editorial AI content
- **Topic 26**: EIN Presswire directory artifact, press release navigation pages

This left **29 technology and domain topics** for downstream analysis.



Technologies Entity Extraction

Identification:

AI technologies and mechanisms were identified from the BERTopic topic assignments.Each resulting topic represents a distinct AI technology, application domain, or mechanism discussed across the corpus, effectively serving as a data-driven taxonomy of AI technologies without requiring a predefined keyword list.

Topic Label	AI Mechanism / Technology Type
AI Hardware and Chip Innovation	Infrastructure Enablement
Generative AI Search Engine Competition	Workflow Redesign
AI Advancements in Healthcare Research	Augmentation
Enterprise AI Data Infrastructure Management	Automation
AI Generated Image Asset Distribution	Content Generation
AI Copyright and Legal Disputes	Regulatory Disruption

Application:

Each article carries both a topic assignment from BERTopic and an industry label from the NER pipeline. Joining these two dimensions produces an industry & technology matrix enabling the analysis to answer not just which industries are impacted by AI, but specifically which AI technologies are driving that impact and in what sentiment direction. This cross-dimensional view forms the basis of the heatmap analysis in the synthesis section.

Organization Entity Extraction

Organization entities were extracted from the cleaned corpus using spaCy's **en_core_web_sm** model. Only the **tok2vec** and **ner** pipeline components were enabled to maximize processing speed.

98% of the articles had **at least 1** organization identified

Noise Filtering Pipeline

- Boundary word filter: entities starting or ending with prepositions ("of", "the", "and") were discarded as malformed extractions
- Stock ticker filter: 2-5 character uppercase sequences matching ticker patterns were removed
- Blocklist filter: ~ 100 generic terms including navigation elements, country names, and boilerplate phrases were explicitly excluded using partial string matching
- AI noise filter: standalone terms like "AI", "LLM", "GPT" were removed as they represent concepts not organizations
- Structural filter: entities containing commas or multiple ampersands were discarded as nav menu fragments
- Duplicates: entities were normalized to title case and deduplicated within each article

Parameter	Value	Rationale
spaCy Model	en_core_web_sm	Lightweight, sufficient for ORG extraction
Text truncation	3,000 characters	Captures lede, reduces compute
Batch size	128	Optimized for CPU
Max entity tokens	5	Filter overly long nav artifacts
Min entity length	3 characters	Remove single letter false positives

Industry Classification Pipeline

01 ORG Volume & Threshold Filtering

Articles **exceeding 10 ORGs**, the 75th percentile were removed as likely stock screener pages, press release directories, and nav-heavy artifacts rather than genuine content.

02 Frequency Analysis

A minimum mention threshold of **5** was applied, retaining only companies with sufficient presence to produce reliable industry-level information.

03 Generative AI Classification

Unique companies meeting the mention threshold were classified into industries using the Gemini API. **gemini-3.1-flash-lite-preview** was used. **Temperature** was set to **0.0** to ensure deterministic, consistent industry assignments across all batches.

04 Prompt Engineering

A structured prompt was used with a constrained **industry list** to ensure consistent, parseable output. An **Unknown** option was included for entities the model could not confidently classify. The model was explicitly instructed to return only valid JSON with no additional text

05 Final Mapping

The completed lookup table was merged back onto the exploded ORG dataframe using a **dictionary map** for a faster merge at scale. Companies not present in the lookup table & those below the minimum mention threshold were assigned **Unknown**.

Sentiment Model

Training Data: The model was trained on the **Financial PhraseBank dataset** (Malo et al., 2014) a collection of **2,264** financial news sentences labeled by domain experts. The **Sentences_AllAgree** subset was used exclusively, meaning only sentences where all annotators reached unanimous agreement on the label maximizing label reliability at the cost of dataset size.
Class distribution was imbalanced: 61% neutral, 25% positive, 13% negative. Stratified splitting was applied to preserve this ratio across all three sets.

roberta-base Model

- **roberta-base** with **124.6M parameters**, **12-layer** transformer encoder fine-tuned for **3-class** sequence classification (negative / neutral / positive)
- **Early stopping** using load_best_model_at_end=True selected the **epoch 4** checkpoint based on validation F1 Macro.
- Performance is high but expected given the Sentences_AllAgree subset represents the least ambiguous, most clearly labeled portion of Financial PhraseBank.

roberta-base Hyperparameter	Value
Epochs	4
Learning Rate	2e-5
Batch Size	16
Weight Decay	0.01
Evaluation Strategy	Per epoch
Best Model Selection	F1 Macro
Mixed Precision	FP16

Epoch	Train Loss	Val Accuracy	F1 Macro
1	0.134	95.8%	0.947
2	0.015	96.2%	0.943
3	0.080	96.9%	0.953
4	**0.062**	**98.3%**	**0.974**

Sentiment Model Inference

Final Corpus Sentiment Distribution

Label	Count	Percentage
Neutral	135,886	69.1%
Positive	54,198	27.6%
Negative	6,545	3.3%

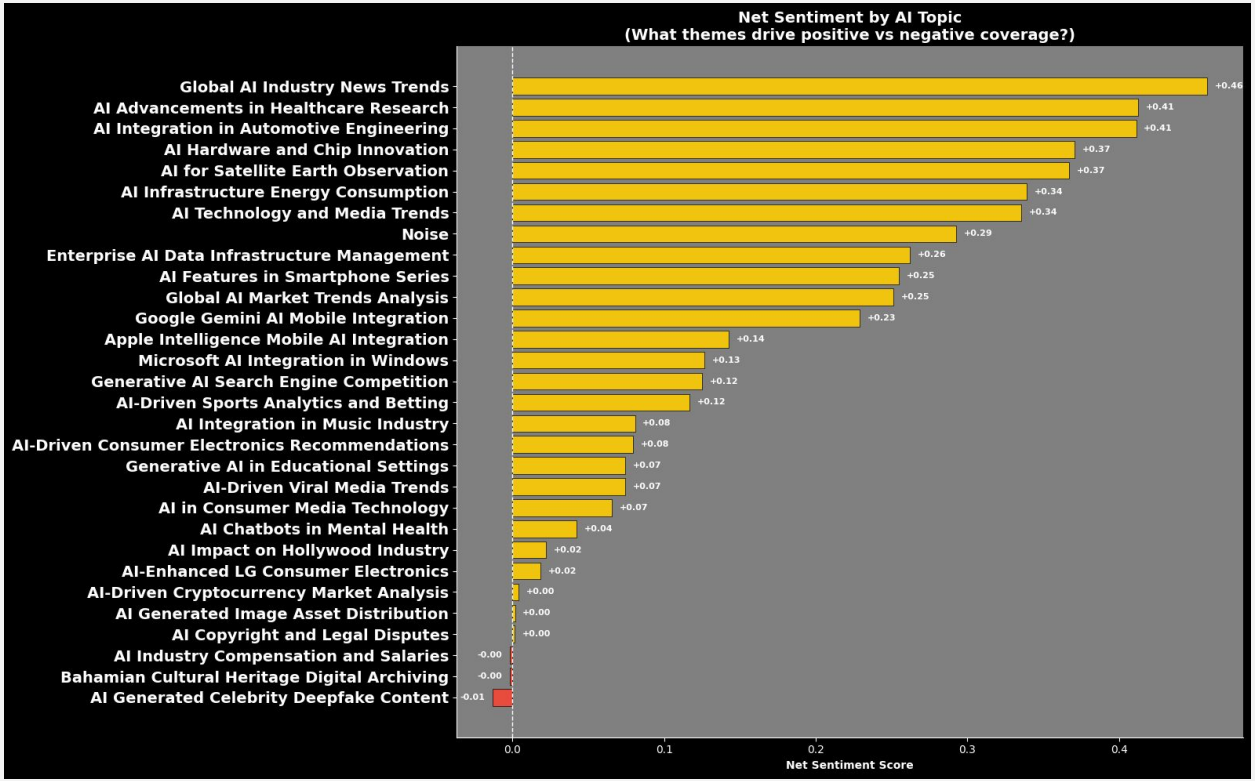
Full Corpus Inference

A token length analysis on a 3,000 article sample revealed that **92% of articles exceed RoBERTa's 512 token limit**, with a mean of 1,707 tokens per article. A sliding window chunking strategy with **stride = 128** was implemented:

- Each article split into overlapping 512-token chunks with a 128-token stride
- Maximum 10 chunks per article
- Each chunk scored independently
- Chunk probabilities aggregated via **inverse-frequency weighted mean** to correct for the neutral-heavy training distribution

Weights: negative x 7.46 | neutral x 1.63 | positive x 3.97

Topic Level Sentiment Analysis

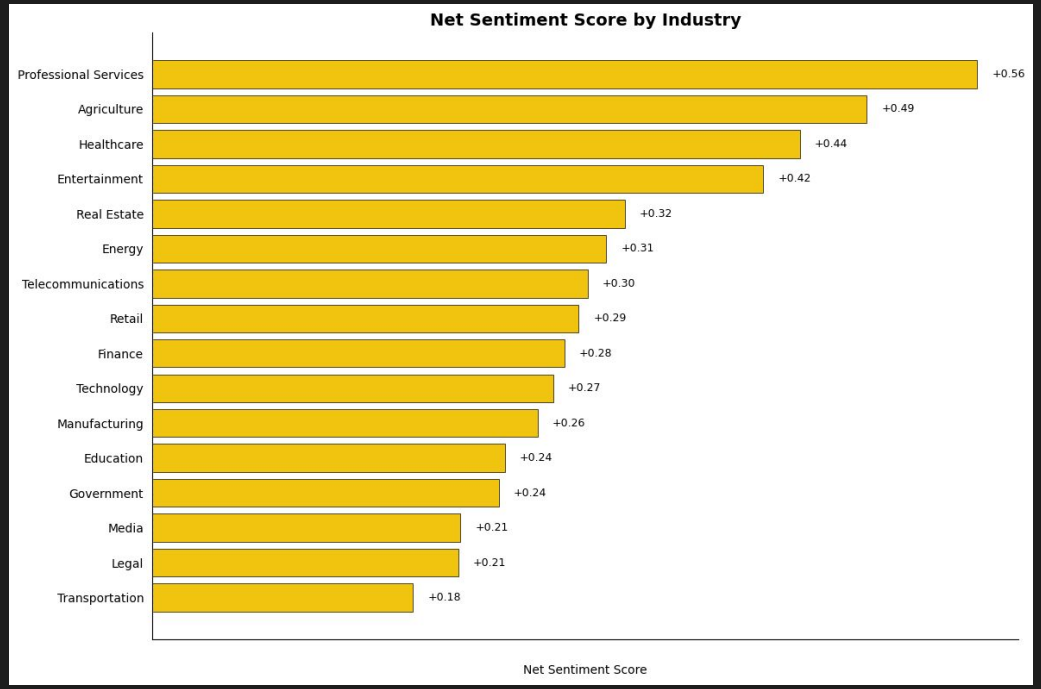


- **Positive Sentiment Leaders:** General **Global AI Industry News (+0.46)**, specialized applications in **Healthcare (+0.41)** and **Automotive (+0.41)** drive the highest net positivity.
- **Infrastructure & Hardware:** Physical components like **Hardware and Chip Innovation (+0.37)** and **Infrastructure Energy Consumption (+0.34)** maintain strong positive sentiment, above specific software integrations.
- **Consumer Tech Disparity:** **Google Gemini AI (+0.23)** holds a higher sentiment score than **Apple Intelligence (+0.14)** and **Microsoft AI (+0.13)** within this dataset.
- **The Neutrality Threshold:** Topics such as **Copyright and Legal Disputes, AI Generated Image Asset Distribution** show no net sentiment, indicating balanced or stagnant public discourse.
- **Negative Sentiment Drivers:** **AI Generated Celebrity Deep Fake Content (-0.01)** is the only topic with a clear negative net score.
- **Labor Concerns:** **AI Industry Compensation and Salaries (-0.00)** sits at the bottom of the spectrum, reflecting a shift toward negative sentiment regarding workforce impact.

Industry Entity Sentiment Analysis

On average, AI coverage across industries runs ~3 positive articles for every negative one

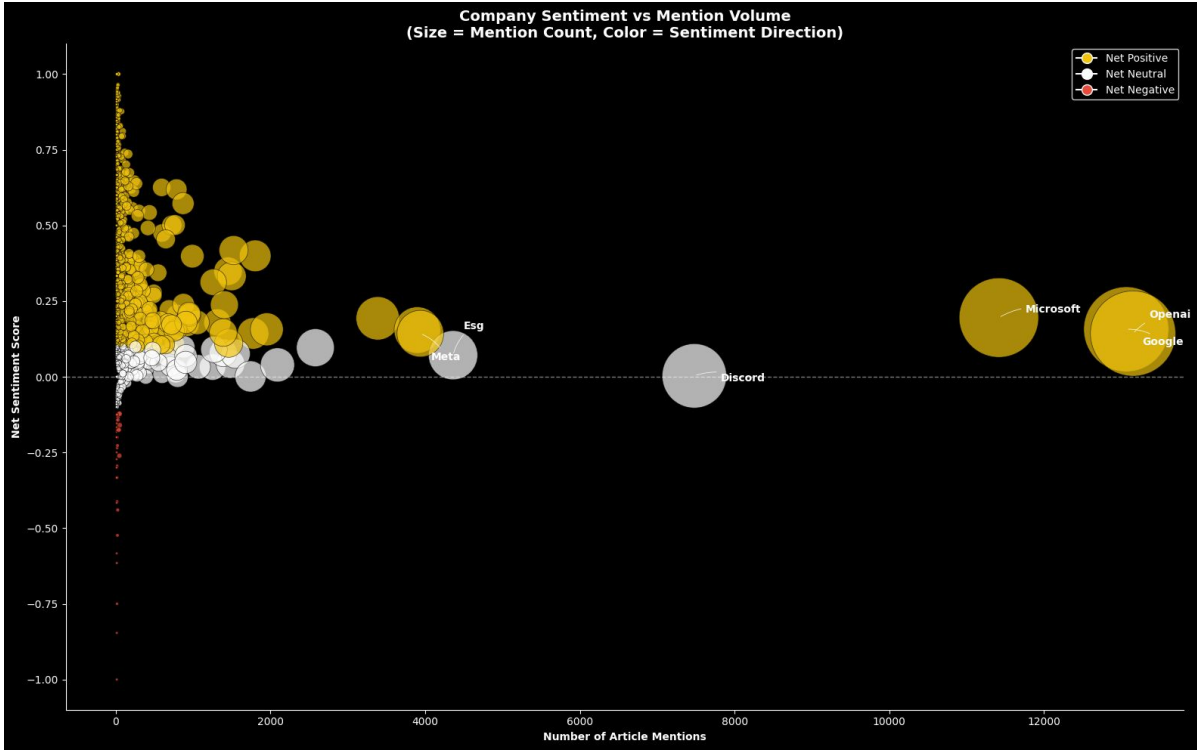
- Professional Services leads with a net sentiment score of +0.56, driven by positive coverage across 16 articles but no negative articles identified.
- Agriculture (+0.49) and Healthcare (+0.44) show strong positive leaning though having smaller article volumes
- Entertainment (+0.42) and Real Estate (+0.32) round out the top five most positively covered industries



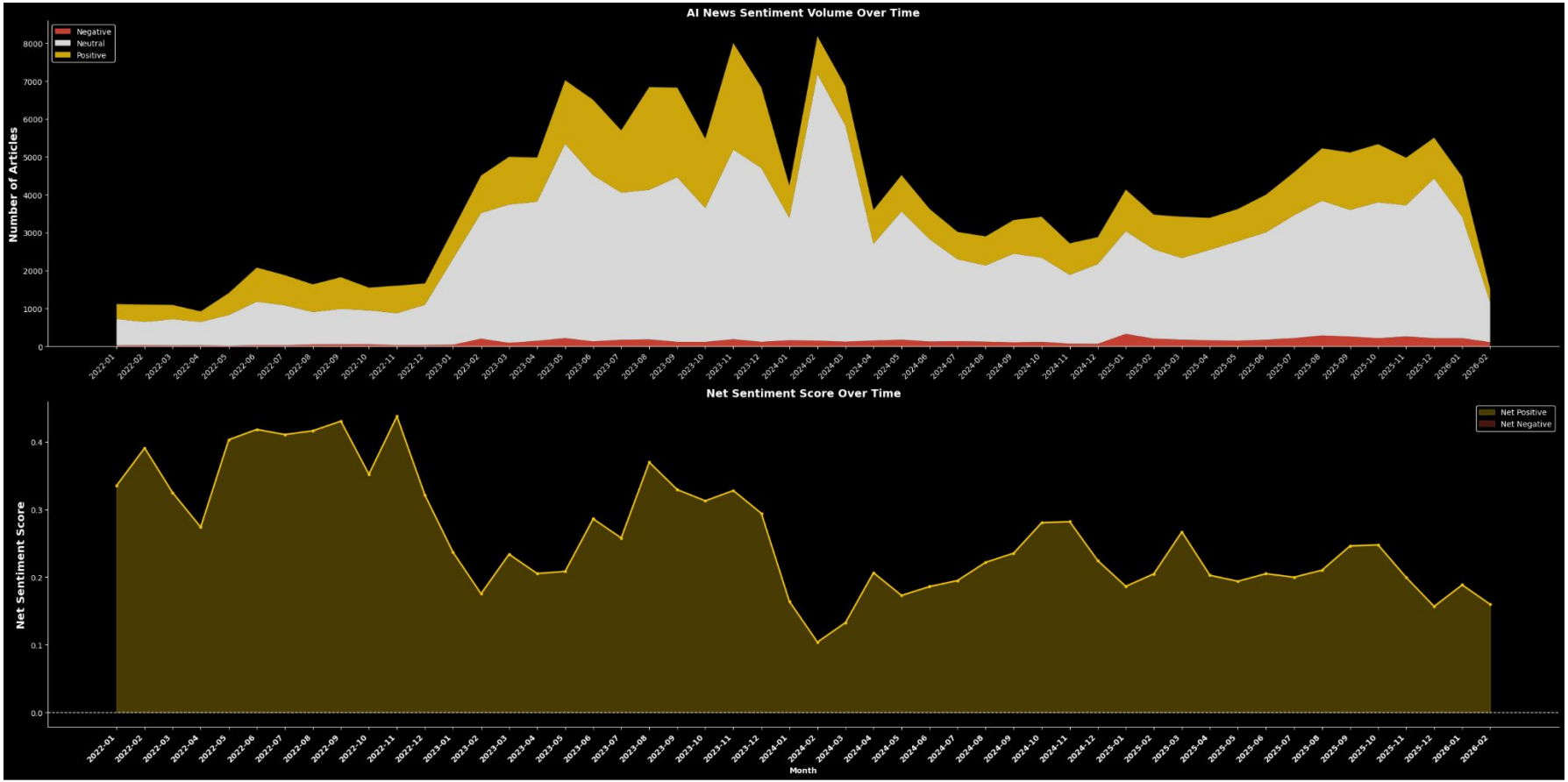
Organization Entity Sentiment Analysis

Are companies positively/negatively impacted?

- **Overall Positive Trend:** Many highly visible companies, such as **OpenAI**, **Google**, and **Microsoft**, generally show a **net positive sentiment** (indicated by yellow bubbles) in AI news coverage. Their large bubble sizes reflect their significant presence and influence in the AI conversations.
- **Varying Sentiments for Other Companies:** While major players lean positive, the chart also reveals a spectrum of sentiments for other companies. Some smaller bubbles are distinctly yellow (positive), indicating strong positive coverage for less frequently mentioned entities, likely for specific innovations or achievements. Conversely, some bubbles are red (negative), suggesting that certain companies face negative coverage, possibly due to controversies or setbacks, even if their overall mention count is lower.



Sentiment Analysis Over Time



Volume Growth: AI news coverage grew dramatically from early 2022 through mid-2024, peaking at roughly 8,000 articles per month. **Sentiment Dilution Over Time:** While article volume increased, net sentiment scores declined from a high of ~0.42 in early 2022 to ~0.15–0.20 range by 2025. **Late 2026 Drop:** The sharp decline in both volume and net sentiment in early 2026 likely reflects the corpus cutoff rather than a genuine trend shift.